Faculty of Artificial Intelligence Engineering
Department of Robotics and Intelligent Systems Engineering

Design and Implementation of a Smart Solar Energy Management and Water Heating System Based on Dual-Axis Tracking and Internet of Things (IoT)

Graduation Project II Report Submitted to the Department of Control Engineering and Robotics in Partial Fulfillment of the Requirements for the Graduation Project Course

Prepared by
Aya Mhd Samer Doumi
Mhd Essam Mhd Nael Kharbotaly

Supervised by
Eng. Marwa AlSafadi

2025-2026

Dedication

I dedicate this work to my family, whose endless support, encouragement, and belief in me have been the foundation of this achievement.
To my supervisor, for her valuable guidance, continuous support, and insightful feedback throughout this project.
To the professors of the University, for their knowledge, mentorship, and dedication.
To the Dean, for his leadership and support in creating a motivating academic environment.
And to everyone who stood by me and supported me during this journey.

Supervisor’s Declaration

I hereby certify that the preparation of this project entitled:

“Design and Implementation of a Smart Solar Energy Management and Water Heating System Based on Dual-Axis Tracking and Internet of Things (IoT)”

prepared by the students:

Aya Mhd Samer Doumi

Mhd Issam Mhd Nael Kharbotaly

has been carried out under my supervision in the Department of Control Engineering and Robotics, Faculty of Engineering, as part of the requirements for obtaining the Bachelor’s degree in Control Engineering and Robotics.

Academic Rank: ....

Name: ....

Signature: ....

Date: ....

Abstract

This project presents the design and implementation of an integrated smart system for solar energy management and water heating, based on dual-axis tracking and Internet of Things (IoT) technologies. The system is developed to enhance energy efficiency, ensure operational reliability, and provide real-time monitoring and control.

The proposed system consists of several interconnected subsystems. A dual-axis solar tracking mechanism dynamically orients the photovoltaic panel toward the maximum solar irradiance, thereby maximizing energy capture and improving overall conversion efficiency throughout the day. The generated energy is stored within a stable and integrated electrical storage system designed to ensure continuous operation and minimize the impact of environmental fluctuations. A closed-loop thermal control system continuously monitors the water temperature inside the tank and regulates the heating process according to a predefined reference value, ensuring optimal thermal performance with minimal energy loss.

In addition, the system incorporates a wireless control and monitoring interface via a smart application, enabling users to remotely adjust temperature settings and observe system status in real time. Safety is enhanced through a water level monitoring system that prevents dry operation, as well as a leakage detection mechanism that triggers immediate protective actions, including water supply isolation and user alerts. A local display interface is also provided to present key operational parameters such as current temperature, target values, and energy status.

The project addresses several key challenges, including inefficient utilization of solar energy, reliance on conventional energy sources for water heating, limited capability for real-time monitoring of temperature and water levels, and the absence of early leak detection systems. Accordingly, the main objectives are to improve energy and water efficiency, reduce dependence on traditional electrical heating, enable intelligent remote monitoring and control, and enhance system safety through automated protection mechanisms.

Overall, the system demonstrates a practical and scalable approach toward sustainable energy utilization and smart resource management.

Keywords

Smart Solar System, Dual-Axis Tracking, Water Heating, Internet of Things (IoT), Thermal Control, Energy Efficiency, Remote Monitoring, Safety System.

الملخص

يقدم هذا المشروع تصميم وتنفيذ منظومة ذكية متكاملة لإدارة الطاقة الشمسية وتسخين المياه، بالاعتماد على تقنية التتبع الشمسي ثنائي المحور وتقنيات إنترنت الأشياء (IoT) ، بهدف تحسين كفاءة استخدام الطاقة، وضمان استمرارية التشغيل، وتوفير مراقبة وتحكم لحظي في النظام.

تتألف المنظومة من عدة أنظمة فرعية مترابطة، حيث تعتمد على آلية تتبع شمسي ثنائي المحور تقوم بتوجيه اللوح الشمسي تلقائياً نحو أعلى شدة إشعاع شمسي، مما يساهم في زيادة القدرة المستخرجة وتحسين كفاءة التحويل الطاقي خلال مختلف أوقات النهار. يتم تخزين الطاقة المنتجة ضمن نظام تخزين كهربائي متكامل يضمن استقرار التغذية واستمرارية عمل النظام حتى في حالات انخفاض الإشعاع الشمسي. كما تتضمن المنظومة نظام تحكم حراري مغلق الحلقة يقوم بمراقبة درجة حرارة المياه داخل الخزان بشكل مستمر، والتحكم بعملية التسخين وفق قيمة مرجعية محددة مسبقاً، بما يحقق الكفاءة الحرارية المطلوبة مع تقليل الفاقد الطاقي.

تدعم المنظومة واجهة تحكم ومراقبة لاسلكية من خلال تطبيق ذكي يتيح للمستخدم ضبط درجة الحرارة ومتابعة الحالة التشغيلية بشكل لحظي وعن بُعد. كما تم تعزيز عناصر الأمان عبر نظام لمراقبة مستوى المياه يمنع التشغيل الجاف، بالإضافة إلى نظام كشف تسرب مائي يعمل على إصدار تنبيهات فورية واتخاذ إجراءات وقائية مثل عزل مصدر المياه. كذلك، توفر المنظومة واجهة عرض محلية تعرض القيم التشغيلية الأساسية مثل درجة الحرارة الحالية، القيمة المطلوبة، وحالة الطاقة.

يعالج المشروع مجموعة من التحديات الأساسية، من أبرزها انخفاض كفاءة استغلال الطاقة الشمسية، والاعتماد على مصادر الطاقة التقليدية في تسخين المياه، وصعوبة المراقبة اللحظية لمستوى المياه ودرجة الحرارة، بالإضافة إلى غياب أنظمة فعالة للكشف المبكر عن التسربات. وعليه، يهدف المشروع إلى تقليل الهدر المائي والطاقي، وتعزيز الاستدامة البيئية، وتمكين التحكم والمراقبة عن بُعد، إضافةً إلى تحسين كفاءة النظام وتوفير آليات حماية تلقائية تضمن سلامة التشغيل.

بشكل عام، يمثل هذا المشروع نموذجاً عملياً لمنظومات الطاقة الذكية التي تدمج بين الطاقة المتجددة وتقنيات التحكم الحديثة، بما يساهم في تحقيق إدارة أكثر كفاءة واستدامة للموارد.

الكلمات المفتاحية

نظام شمسي ذكي، تتبع ثنائي المحور، تسخين المياه، إنترنت الأشياء، التحكم الحراري، كفاءة الطاقة، المراقبة عن بعد، نظام الأمان.

Table of Contents

List of Tables

List of Figure

Abbreviation	Full Term
ESP	Espressif Microcontroller Platform
IoT	Internet of Things
LDR	Light Dependent Resistor
I/O	Input / Output
DC	Direct Current
GPIO	General Purpose Input/Output
I2C	Inter-Integrated Circuit
SDA	Serial Data Line
SCL	Serial Clock Line
KPIs	Key Performance Indicators
PID	Proportional-Integral-Derivative

List of Abbreviations

List of Symbols

Symbol	Description
Q	Heat energy
M	Mass of water
C	Specific heat capacity
ΔT	Temperature difference
T	Temperature
T	Time
P	Power
H	System efficiency
V	Voltage
I	Current
θ	Angular position (servo angle)

Chapter 1

General Introduction

1.1 Project Overview and Background

In recent years, there has been a growing global interest in utilizing renewable energy sources, particularly solar energy, due to its sustainability and environmental benefits. However, one of the major limitations of conventional solar systems is their relatively low efficiency, as solar panels are typically fixed at a constant angle, resulting in significant energy loss throughout the day.

To address this issue, solar tracking systems have been developed to continuously orient solar panels toward the sun using light sensors and motorized mechanisms, thereby maximizing solar radiation capture and improving overall system performance. Previous studies have demonstrated that solar tracking systems significantly increase energy output compared to fixed solar panels [1].

Based on this concept, the development of this project began with a single-axis solar tracking system implemented during the semester project, which was later enhanced into a dual-axis tracking system in Graduation Project 1 to achieve higher tracking accuracy. In the current stage, the system has been further extended to include a smart solar water heating application, integrated with Internet of Things (IoT) technology for real-time monitoring and remote control.

The proposed system is built around an ESP32 microcontroller, which integrates various sensors and actuators into a unified platform. Light Dependent Resistors (LDRs) are used to detect the direction of solar radiation and control the movement of the solar panel using servo motors. In addition, a DS18B20 digital temperature sensor is employed for accurate water temperature measurement, as it has been proven to provide reliable and stable readings in low-cost thermal monitoring applications [2]. Furthermore, the system incorporates water level and leakage detection sensors to ensure safe operation, as studies have shown that such systems enhance operational safety and reduce potential risks [3][4].

Accordingly, this project presents an integrated smart system that combines solar tracking, thermal control, and safety mechanisms within a single platform, aiming to improve solar energy utilization efficiency and provide a reliable solution for solar water heating applications.

1.2 Problem Statement and Importance

Despite the widespread adoption of solar water heating systems, many existing solutions still suffer from significant limitations in terms of efficiency and operational safety, which restrict their practical effectiveness. One of the primary challenges is the reliance on fixed solar panels that do not adapt to the sun's movement throughout the day, resulting in reduced solar energy capture and overall system efficiency. Previous studies have demonstrated that solar tracking systems can significantly enhance energy output compared to fixed systems [1].

In addition, conventional heating systems often lack precise thermal control mechanisms, where the heating element continues to operate without considering the required temperature limit, leading to unnecessary energy consumption. Furthermore, the absence of accurate temperature measurement negatively affects system performance, whereas studies have shown that digital sensors such as the DS18B20 provide reliable and stable temperature readings in thermal monitoring applications [2].

From a safety perspective, these systems present several operational risks. Heating may occur even when the water level is low or when the tank is empty, potentially causing system damage or hazardous conditions. Moreover, the lack of effective leakage detection mechanisms makes it difficult to identify faults at an early stage, although conductivity-based detection systems have been proven to provide a simple and efficient solution for detecting water leakage [4]. Similarly, water level monitoring systems have been shown to improve system management and prevent overflow or shortage conditions [3].

Another critical limitation is the absence of intelligent monitoring capabilities, which prevents users from tracking system performance in real time, particularly when the system is installed in remote or elevated locations. This lack of monitoring reduces system reliability and increases the likelihood of undetected failures.

Accordingly, there is a clear need for an integrated solution that enhances solar energy utilization, ensures precise thermal control, improves operational safety, and enables real-time monitoring. Therefore, this project proposes a smart system that incorporates a dual-axis solar tracking mechanism to maximize solar energy capture, combined with a precise thermal control system, supported by reliable sensors and safety mechanisms including water level monitoring and leakage detection. In addition, an Internet of Things (IoT)-based platform is implemented to allow remote monitoring and control of the system in an efficient and user-friendly manner.

1.3 Objectives

The project objectives are defined based on measurable performance indicators to ensure a systematic evaluation of system efficiency, safety, and reliability, following the SMART criteria (Specific, Measurable, Achievable, Relevant, and Time-bound).

These objectives are based on well-established principles in previous studies, including solar tracking efficiency improvement [1], accurate temperature measurement [2], automated water level control [3], and leakage detection systems [4].

1.3.3 Main Objective

The main objective of this project is to design and implement an integrated smart system for solar energy management and water heating, based on dual-axis solar tracking and Internet of Things (IoT) technologies, in order to enhance solar energy utilization, reduce energy losses, and ensure safe and stable operation according to user-defined parameters.

1.3.2 Specific Objectives

1. **Solar Tracking Performance**

   To achieve dual-axis solar tracking with an angular accuracy of approximately ±5° and a response time of 1–2 seconds, aiming to improve solar energy capture efficiency by an estimated 20–40% compared to fixed systems [1].

2. **Thermal Heating Control**

   To heat water within a temperature range of 40–60°C, while maintaining temperature stability within ±2°C around the setpoint, using a DS18B20 digital temperature sensor known for its reliability and stability [2].

3. **Real-Time Monitoring and IoT Control**

   To provide real-time monitoring with an update rate of ≤ 1 second, enabling remote control via a mobile application, along with local data display using an LCD interface.

4. **Safety and Protection System**

To ensure water leakage detection within ≤ 1 second using conductivity-based methods [4], and to prevent heater operation in case of low or absent water level based on monitoring techniques [3], in addition to automatic shutdown at excessive temperatures.

5. **Energy Efficiency Optimization**

To reduce electrical energy consumption by an estimated 20–30% through efficient solar tracking and eliminating unnecessary heating operation.

6. **System Reliability and Stability**

To ensure continuous system operation (24/7 monitoring) while minimizing oscillations in tracking movement and heater operation.

1.3.3 Mathematical Basis for Performance Evaluation

The thermal performance of the system is evaluated based on the amount of heat energy required to raise the water temperature, which is expressed as :

$$Q = m * c * \Delta T$$

Where

- Q is the required thermal energy
- m is the mass of water
- c is the specific heat capacity
- ΔT is the temperature difference between the initial and target temperatures

The expected heating time is calculated based on the heater power using the following relation:

$$t = Q \setminus P$$

Where

- t represents the heating time
- P is the heater power

Based on these equations, the heating duration typically ranges between 2 to 6 hours depending on the water volume, temperature difference, heater power, and solar irradiance conditions, while also considering system thermal losses.

1.4 Definitions and Basic Concepts

This project is based on a set of fundamental concepts related to solar energy systems, control engineering, and Internet of Things (IoT) technologies, which form the theoretical foundation for the design and implementation of the proposed system. Understanding these concepts is essential for analyzing system behavior and evaluating its performance.

Solar energy refers to the energy derived from solar radiation, which can be converted into thermal or electrical energy and is widely used in renewable energy applications, particularly for water heating systems.

Solar tracking systems are used to automatically orient solar panels toward the sun in order to maximize solar radiation capture. Studies have shown that dual-axis tracking systems significantly improve energy efficiency compared to fixed systems [1].

In this project, the tracking mechanism is based on **Light Dependent Resistors (LDRs)**, which are sensors whose resistance varies according to light intensity, enabling the detection of solar direction.

The movement of the solar panel is controlled using **servo motors**, which provide precise angular positioning.

The system is managed by an **ESP32 microcontroller**, which offers both processing capabilities and wireless communication, enabling the integration of **Internet of Things (IoT)** technologies for real-time monitoring and remote control.

For thermal regulation, the system utilizes a **DS18B20 digital temperature sensor**, which provides accurate and stable temperature measurements and has been proven suitable for low-cost thermal monitoring applications [2].

The heating process is controlled through a **thermal control system**, which maintains the water temperature within a predefined range based on the user-defined **setpoint**.

To ensure safe operation, the system incorporates a **water level sensor** to monitor the tank condition and prevent unsafe operation, as water level monitoring systems have been shown to improve system management and safety [3].

In addition, a **leakage detection system** is implemented based on electrical conductivity principles, where the presence of water bridges conductive electrodes, enabling early detection of leaks [4].

Finally, system performance is evaluated based on **system efficiency**, which represents the ability of the system to convert available energy into useful output while minimizing energy losses.

1.5 Scope and Technical Requirements

1.5.1 Project Scope

This project aims to design and implement a smart embedded system for solar energy management and water heating, specifically targeted for residential applications. The system integrates dual-axis solar tracking, thermal control, and Internet of Things (IoT) technologies into a unified platform.

The system includes a dual-axis solar tracking mechanism to maximize solar radiation capture, along with a thermal control system based on a user-defined setpoint, where the heating process is automatically controlled (ON/OFF) according to the desired temperature. In addition, the system provides real-time monitoring through a mobile application connected via Wi-Fi, as well as local data visualization using an LCD display.

The system incorporates multiple sensors to measure light intensity, water temperature, water level, electrical voltage, and leakage conditions. These components are managed by an ESP32 microcontroller, which serves as the central processing unit responsible for decision-making and controlling actuators such as servo motors, the heating element, and the water pump.

Furthermore, the system is designed with a strong focus on safety by preventing operation under unsafe conditions, such as low or empty water levels, and by enabling early leakage detection, in addition to implementing thermal shutdown at the desired temperature.

However, this project does not include artificial intelligence or machine learning techniques, does not support grid connection, and is not intended for large-scale or industrial applications. It also excludes advanced features such as weather prediction or intelligent load optimization.

Accordingly, the system can be defined as a smart embedded energy management system for residential use, focusing on efficiency, safety, and real-time monitoring without introducing unnecessary complexity.

1.5.2 Technical Requirements

The technical requirements of the system are defined based on the intended performance objectives and include mechanical, thermal, electrical, and control aspects. The following table summarizes the key specifications of the system:

Table 1.1:Technical Specifications of the System

Section	Technical Specification	Value / Requirement	Verification Method
Solar Tracking System	Tracking Accuracy	Approximately ±5°	Practical testing using LDR sensors
Solar Tracking System	Energy Gain vs Fixed Panel	20–40% improvement	Performance comparison with fixed panel (Literature-based)
Solar Tracking System	Tracking Mechanism	Dual-axis (2 Servo Motors + 4 LDR)	Hardware inspection
Solar Tracking System	Sensor Type	4× LDR (Analog Sensors)	Signal testing
Thermal Control System	Temperature Measurement Range (DS18B20)	-55°C to +125°C	Datasheet
Thermal Control System	Measurement Accuracy	±0.5°C (within -10°C to +85°C)	Datasheet
Thermal Control System	Control Logic	T < Setpoint → ON / T ≥ Setpoint → OFF	Software testing
Thermal Control System	Control Type	Closed-loop control	System analysis
Water Management System	Water Level Sensor	Analog (0–4095 ADC)	Reading verification

Water Management System	Pump Operation	Automatic ON at low level	Functional testing
Water Management System	Leak Detection	Conductivity-based detection	Water testing
Control Unit	Microcontroller	ESP32 (Dual-core + Wi-Fi)	Datasheet + code verification
IoT System	Communication Type	Wi-Fi (Local Network)	Connectivity testing
IoT System	Data Update Rate	1–5 seconds	Application monitoring
IoT System	Application Features	Monitoring + Control + Alerts	App testing
Safety System	Dry-run Protection	Heater OFF when water level is LOW	Safety testing
Safety System	Thermal Control Safety	Heater OFF at Setpoint	Functional testing
Power System	Operating Voltage	5V (control) / 12V (heater)	Electrical measurement
Power System	ESP32 Power Consumption	~80–260 mA	Datasheet
System Performance	Overall Efficiency	Up to 40% improvement using tracking	Literature comparison

1.5.3 Analytical Notes

The DS18B20 temperature sensor provides a measurement accuracy of approximately ±0.5°C within its operating range, which is sufficient for ON/OFF thermal control systems, resulting in a practical temperature stability of around ±2°C.

The solar tracking accuracy of ±5° is appropriate for LDR-based systems, offering a balance between system simplicity and performance, while achieving an energy gain of approximately 20–40% compared to fixed systems.

The system operates using a 12V battery charged by solar energy, while the control circuitry operates at 5V, ensuring proper separation between power and control levels.

The system is designed for continuous operation (24/7 monitoring) with stable performance by minimizing oscillations in tracking movement and heating control through threshold-based decision logic.

The system design is based on integrating proven technologies from previous studies, including solar tracking systems [1], digital temperature sensing [2], automated water level control [3], and leakage detection mechanisms [4].

1.6 Project Contribution

This project presents a practical contribution through the development of an integrated system that combines solar tracking, thermal control, and Internet of Things (IoT) technologies into a unified platform for residential applications. Unlike conventional systems that rely on separate functionalities, the proposed system integrates energy harvesting, heating control, and real-time monitoring within a single solution.

The project has evolved progressively through multiple stages. Initially, a single-axis solar tracking system was implemented during the semester project, followed by a dual-axis tracking system in Graduation Project 1 to improve tracking accuracy. In the current stage, the system has been extended to include a complete smart water heating application, incorporating user-defined temperature control and IoT-based remote monitoring.

The main contribution of this project lies in the effective integration of solar tracking and thermal heating within a single system, which enhances solar energy utilization and reduces reliance on conventional electrical energy. In addition, the system improves operational safety by incorporating protection mechanisms such as water leakage detection and dry-run protection, minimizing potential risks.

Furthermore, the system provides an interactive user interface that enables remote monitoring and control of water temperature through a mobile application, along with real-time alerts in case of abnormal conditions such as leakage or low water levels. This integration of monitoring and control contributes to reducing unnecessary heater operation, thereby minimizing energy waste and improving overall system efficiency.

Accordingly, this project represents an advanced step toward the development of smart solar water heating systems that combine efficiency, safety, and usability within a compact and cost-effective solution.

Unlike previous studies that focused on individual subsystems separately [1]–[4], this project presents an integrated system that combines solar tracking, thermal control, IoT monitoring, and safety mechanisms within a unified platform.

1.7 Design Methodology

A structured design methodology was adopted in this project, combining theoretical analysis with practical implementation to ensure the development of an efficient and integrated system that meets the defined performance requirements. Initially, a comprehensive theoretical study was conducted based on relevant research in solar tracking systems, thermal sensing, and control systems, in order to identify suitable approaches and select appropriate components.

Following the theoretical phase, the system was designed and implemented by dividing it into several subsystems, including the solar tracking system, thermal heating system, IoT monitoring system, and safety and protection system. This modular approach facilitated system development, allowing each subsystem to be analyzed and optimized independently before full system integration.

Component selection was based on criteria such as accuracy, efficiency, cost, and ease of integration. LDR sensors were used for solar tracking, servo motors for precise positioning, and a DS18B20 sensor for accurate and stable temperature measurement. The ESP32 microcontroller was selected due to its processing capabilities and built-in wireless communication, enabling the implementation of a smart and connected control system.

The control strategy is based on a comparison-driven decision logic. The solar panel orientation is determined by comparing LDR sensor readings, while the heating process is controlled based on a predefined temperature setpoint. The heater is activated when the temperature falls below the setpoint and deactivated once the desired temperature is reached. In addition, safety mechanisms are triggered immediately in abnormal conditions such as water leakage or low water level.

The system was implemented using Arduino IDE for programming the ESP32, while Android Studio was used to develop the mobile application for remote monitoring and control. After implementation, the system was tested experimentally to evaluate its performance and stability. The results demonstrated correct system response under different operating conditions, achieving the intended functionality efficiently.

Table 1.2: System Components and Justification

Component	Reason for Selection
DS18B20	Provides high accuracy, measurement stability, easy integration, and suitability for smart control systems.
MG995 Servo Motor	Offers high torque, suitable speed, durability, ease of control, and wide availability.
LCD 16×2	Provides clear display, easy interfacing, real-time data updates, low cost, and ease of use.
LDR Sensor	Enables intelligent light tracking, simple installation, low cost, and sufficient accuracy for the application.
DC Voltage Meter	Ensures accurate and stable voltage monitoring, battery protection, and clear data visualization.
Water Level Sensor	Enhances system safety, prevents dry operation, and provides accurate water level monitoring.
Water Leakage Sensor	Enables instant leak detection, automatic protection, smart alerts, and fast response.
Water Valve	Ensures operational safety, protects system components, and provides immediate response to abnormal conditions.
BD243 Transistor	Allows safe control of the 12V heater, supports high current, ensures long-term stability, and integrates easily with microcontrollers.
Voltage Regulator	Provides stable voltage supply, protects electronic components, and ensures stable system performance.
Buzzer	Provides instant audible alerts, real-time response, and easy integration into the system.
ESP32 Microcontroller	Offers built-in Wi-Fi connectivity, high processing capability, multiple I/O pins, and flexible programming support.

The adopted design methodology is based on concepts validated in the literature, including solar tracking techniques [1], temperature sensing systems [2], water level control strategies [3], and safety mechanisms such as leakage detection [4].

1.8 Report Structure

This report is organized into five main chapters, in addition to appendices and references, ensuring a structured and systematic presentation of the project.

Chapter 1: General Introduction provides an overview of the project, its importance, objectives, basic concepts, and the design methodology adopted.

Chapter 2: Literature Review presents relevant studies and previous work related to the project, including a comparative analysis and identification of the research gap.

Chapter 3: System Analysis and Design describes the overall system architecture, components, and the mathematical modeling and engineering analysis used in the design process.

Chapter 4: Implementation and Testing focuses on the practical implementation of the system, along with experimental results, performance evaluation, and system validation.

Chapter 5: Conclusion and Recommendations summarizes the main achievements of the project, discusses the challenges encountered, and provides suggestions for future improvements and recommendations.

In addition, the report includes **appendices** containing detailed system designs, source code, and user manual, as well as a **references section** listing all scientific sources used in this work.

1.9 Chapter Conclusion

In this chapter, a comprehensive overview of the project has been presented, including its general concept, significance, problem statement, and objectives. The fundamental concepts and design methodology were also discussed, along with the project scope and technical requirements. Furthermore, the contribution of the project to the development of smart solar water heating systems has been highlighted.

These elements establish the theoretical and organizational foundation of the project and prepare for the next chapter, which focuses on reviewing existing studies and solutions in the field, in order to analyze them and utilize their findings in the development of the proposed system.

Chapter 2

Literature Review

2.1 Previous Studies in the Field

Recent advancements in solar energy and smart control systems have focused on improving energy efficiency, measurement accuracy, and system safety.

Study [1] presented a practical solar tracking system based on light sensors and mechanical actuation, demonstrating that solar tracking can increase power output by more than 30% compared to fixed systems by maintaining an optimal angle of solar radiation incidence.

In terms of thermal measurement, study [2] evaluated the DS18B20 sensor and showed strong agreement with reference sensors, with minimal deviation reaching ±0.08°C under stable conditions, confirming its suitability for accurate and stable temperature monitoring.

In the field of water management, study [3] proposed an Arduino-based automatic water level control system, demonstrating effective prevention of dry-run conditions and improved water management efficiency through automated sensor-based control.

Regarding safety systems, study [4] described a conductivity-based water leakage detection mechanism, where leakage is detected through the formation of an electrical path between electrodes, enabling fast and reliable response.

Overall, these studies focused on developing separate subsystems such as solar tracking [1], temperature sensing [2], water management [3], and safety systems [4], without integrating them into a unified smart system capable of real-time monitoring and control.

Table 2.1: Summary of Previous Studies

Study	Focus Area	Key Contribution	Limitation
[1] Solar Tracking System	Solar Energy	Increased power output >30%	No integration with heating or IoT
[2] DS18B20 Study	Temperature Measurement	High accuracy (±0.08°C)	Limited to sensing only
[3] Water Level Control	Water Management	Automatic pump control, dry-run protection	No energy system integration
[4] Leakage Detection	Safety System	Fast and reliable leak detection	Standalone function

2.2 Comparison with Existing Systems

Based on the reviewed studies, it is evident that most existing solutions focus on developing specific functionalities within isolated systems, without achieving full integration between different system components. Study [1] focused on improving energy efficiency through solar tracking, while study [2] addressed accurate and stable temperature measurement. In parallel, study [3] concentrated on water level management and dry-run prevention, whereas study [4] provided an effective solution for water leakage detection.

Although these solutions are effective within their respective domains, they lack integration between solar tracking, thermal control, water management, and safety systems. Furthermore, they do not support real-time monitoring or remote control capabilities.

In contrast, the proposed system in this project offers a fully integrated solution that combines all these functionalities within a single platform. It incorporates dual-axis solar tracking to enhance energy efficiency [1], accurate temperature measurement using the DS18B20 sensor [2], automated water level control [3], and a leakage detection mechanism to ensure system safety [4].

Moreover, the system integrates Internet of Things (IoT) technologies, enabling real-time monitoring and remote control through a mobile application, along with local data visualization using an LCD display. This significantly enhances system usability and operational efficiency.

Therefore, the proposed system can be considered more comprehensive and advanced compared to existing solutions, as it does not focus on a single subsystem but rather provides an integrated approach combining efficiency, safety, and smart monitoring within one system.

Table 2.2: Comparison Between Existing Systems and Proposed System

Feature / System	**[1] Solar Tracking**	**[2] Temperature Sensor**	**[3] Water Level Control**	**[4] Leak Detection**	**Proposed System**
Solar Tracking	✓	✕	✕	✕	✓ (Dual-axis)
Temperature Measurement	✕	✓	✕	✕	✓
Water Level Control	✕	✕	✓	✕	✓

Leak Detection	✗	✗	✗	✓	✓
IoT Monitoring	✗	✗	✗	✗	✓
Remote Control	✗	✗	✗	✗	✓
LCD Display	✗	✗	✓ (limited)	✗	✓
System Integration	✗	✗	✗	✗	✓

2.3 Research Gap

Despite the significant advancements in solar energy systems and smart control technologies, previous studies have primarily focused on developing individual subsystems rather than integrated solutions. Solar tracking systems have demonstrated their ability to improve energy efficiency by maintaining an optimal angle of solar radiation incidence [1], while digital temperature sensors such as the DS18B20 have shown high accuracy and stability in temperature measurement [2].

In parallel, automated water level control systems have provided effective solutions for preventing dry-run conditions and improving water management [3], whereas leakage detection systems have enabled fast and reliable fault detection based on electrical conductivity principles [4].

However, despite their importance, these studies have addressed each subsystem independently, without achieving full integration between solar tracking, thermal control, water management, and safety systems within a single platform.

Furthermore, there is a clear lack of integration of Internet of Things (IoT) technologies in these systems, as most existing solutions do not support real-time monitoring or remote control, limiting their practical usability and user interaction.

Therefore, a clear research gap exists in the development of a fully integrated smart system that combines solar tracking, thermal control, water management, safety mechanisms, and IoT-based monitoring and control, aiming to improve system efficiency, reduce energy consumption, and ensure safe and reliable operation.

2.4 Theoretical Framework

The proposed project is based on fundamental theoretical concepts in control systems, solar energy, and Internet of Things (IoT) technologies. A simple and effective control methodology based on ON/OFF control is adopted, which is widely used in thermal systems due to its simplicity and reliability [2].

In the solar tracking subsystem, the theoretical approach is based on comparing light intensity using multiple LDR sensors placed around the solar panel. The system determines the optimal direction by comparing sensor readings and orienting the panel toward the highest light intensity, thereby improving energy capture efficiency, as demonstrated in light-based solar tracking systems [1].

From a thermal perspective, the system relies on basic heat transfer principles, where the heating process is controlled based on the difference between the current temperature and the desired setpoint. This is supported by a simplified energy model that relates thermal energy to heating time [5].

In terms of system components, the design is based on the integration of sensors, actuators, and microcontrollers. Sensors (such as LDR, DS18B20, water level, and leakage sensors) provide real-time data, while actuators (such as servo motors, heater, and pump) execute control actions. The ESP32 microcontroller serves as the central processing unit that coordinates all system operations.

Additionally, the system incorporates IoT concepts by enabling wireless communication between the system and a mobile application via Wi-Fi, allowing real-time monitoring and remote control, which enhances system flexibility and usability.

Overall, the theoretical framework combines simplicity and effectiveness by employing appropriate control strategies and models that align with the system's design, ensuring a balance between performance, ease of implementation, and reliability.

2.5 Chapter Conclusion

This chapter reviewed a set of previous studies related to solar tracking systems, temperature sensing, water management, and safety mechanisms. The findings showed that solar tracking systems significantly improve energy utilization and increase power output [1], while digital sensors such as the DS18B20 provide high accuracy and stability in temperature measurement [2]. In addition, water level control systems effectively enhance water management and prevent dry-run conditions [3], whereas leakage detection systems enable fast and reliable responses to potential faults [4].

Despite these contributions, most studies focused on developing individual subsystems independently, without achieving full integration between system components. Furthermore, many of these solutions lack the integration of Internet of Things (IoT) technologies, limiting real-time monitoring and remote control capabilities.

Accordingly, there is a clear need for a fully integrated system that combines these functionalities within a unified platform, which is addressed in this project. Therefore, the next chapter focuses on the analysis and design of the proposed system based on the concepts and findings derived from the reviewed studies.

Chapter 3

System Analysis and Design

3.1 System Overview and Architecture

The proposed system represents an integrated solution for solar energy management and water heating, combining solar tracking, thermal control, water management, and safety mechanisms within a single platform based on an ESP32 microcontroller. This integration aims to improve solar energy utilization efficiency and ensure safe and reliable system operation, based on well-established principles such as solar tracking systems [1], accurate temperature sensing [2], water level control [3], and leakage detection mechanisms [4].

The system consists of multiple sensors that provide the required input data for decision-making. Four Light Dependent Resistors (LDRs) are used to detect solar radiation direction, while a DS18B20 sensor is employed for accurate water temperature measurement. In addition, a water level sensor monitors the tank condition, and a water leakage sensor detects abnormal conditions.

Based on these inputs, the ESP32 microcontroller executes a control algorithm to manage the system actuators. These include MG995 servo motors for dual-axis solar tracking, a heating element for temperature control based on the user-defined setpoint, a water pump and solenoid valve for water management, and a buzzer for safety alerts.

The system also provides two interfaces for monitoring and user interaction. A local LCD display presents real-time system data, including current temperature, battery voltage, and the user-defined temperature setpoint. Additionally, a mobile application enables remote monitoring and allows the user to set the desired temperature.

The system operates continuously in real time, where sensor data are periodically acquired, processed, and used to make control decisions, ensuring fast response to changing operating conditions.

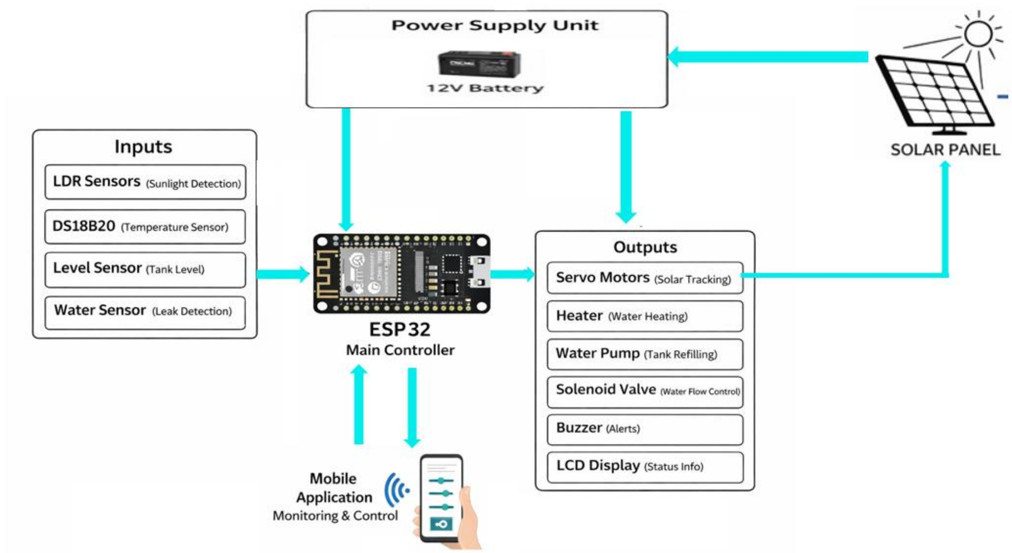

Figure 3.1: Block Diagram of the Proposed System

3.2 System Components

The proposed system consists of an integrated set of electronic and mechanical components that work together to achieve solar tracking, thermal control, water management, and safety functions. The selection of these components was based on criteria such as accuracy, reliability, efficiency, and ease of integration, following well-established concepts in previous studies, including solar tracking systems [1], accurate temperature sensing [2], water level control [3], and leakage detection mechanisms [4].

The system includes various sensors that provide essential input data for decision-making, such as LDR sensors for detecting solar radiation direction, a DS18B20 sensor for accurate and stable temperature measurement, as well as water level and leakage sensors to ensure safe operation. The ESP32 microcontroller serves as the central control unit, responsible for processing data and executing control algorithms.

At the actuator level, MG995 servo motors are used to enable dual-axis solar tracking, while a heating element controls the water temperature based on the user-defined setpoint. Additionally, a water pump and solenoid valve manage water flow, and a buzzer provides safety alerts during abnormal conditions. An LCD display is used for local data visualization, and a voltage monitoring system ensures stable power supply.

This integration of components results in a balanced system that combines high performance, safety, and user-friendly operation, making it suitable for smart residential applications.

Table 3.1: Detailed System Components and Selection Justification

Component	Category	Function	Reason for Selection
LDR (×4)	Sensor	Detect solar light intensity and direction	Low cost, simple design, sufficient accuracy for tracking [1]
DS18B20	Sensor	Measure water temperature	High accuracy, stability, digital output [2]
Water Level Sensor	Sensor	Monitor water level in tank	Prevent dry-run operation, ensure system safety [3]
Water Leakage Sensor	Sensor	Detect water leakage	Fast detection using conductivity principle [4]
DC Voltage Sensor	Sensor	Monitor battery voltage	Ensures stable power and system protection
ESP32	Control Unit	Process data and control system	Built-in Wi-Fi, high performance, flexibility

MG995 Servo Motor (×2)	Actuator	Control solar panel position	High torque, durability, suitable for tracking
Heater (12V)	Actuator	Heat water to setpoint	Efficient and easy control
Water Pump	Actuator	Transfer water between tanks	Reliable fluid control
Solenoid Valve	Actuator	Control water flow	Fast response and safe operation
Buzzer	Actuator	Provide alert signals	Immediate warning in fault conditions
LCD 16×2	Interface	Display system data	Real-time monitoring, easy interface

3.3 Control Strategy

The control strategy of the proposed system is based on a loop-based control algorithm implemented within the ESP32 microcontroller, where sensor data are continuously read, processed, and used to make real-time decisions. The strategy is designed to achieve four main functions: solar tracking, thermal control, water management, and safety protection, based on well-established principles from previous studies such as solar tracking systems [1], accurate temperature sensing [2], water level control [3], and leakage detection mechanisms [4].

The solar tracking system operates as a parallel process, where LDR sensor readings are continuously compared to determine the direction of maximum light intensity, and the servo motors are adjusted accordingly to orient the solar panel toward the sunlight. This approach improves solar energy capture efficiency, as demonstrated in solar tracking systems [1].

Within the main control flow, the system begins by reading all sensors, including battery voltage, temperature, water level, and leakage status. The battery voltage is used for monitoring purposes only and is displayed on both the LCD and the mobile application without directly affecting system operation decisions.

The system then checks for leakage conditions. If leakage is detected, immediate safety actions are triggered, including activating the buzzer, turning off the heater, controlling the pump and solenoid valve, and sending an alert to the mobile application. This mechanism is based on the conductivity principle used in leakage detection systems [4].

If no leakage is detected, the system proceeds to thermal control. The user-defined temperature setpoint is received from the mobile application and compared with the current water temperature measured using the DS18B20 sensor. If the measured temperature is below the setpoint, the heater is activated; otherwise, it is turned off, ensuring stable and accurate temperature regulation [2].

Additionally, the system verifies the water level before activating the heater. If the water level is low, the heater is turned off and alert mechanisms are activated to prevent dry-run conditions, following the principles of water level control systems [3].

At the end of each operation cycle, the LCD is updated and data are transmitted to the mobile application, including the current temperature, desired temperature, and battery voltage.

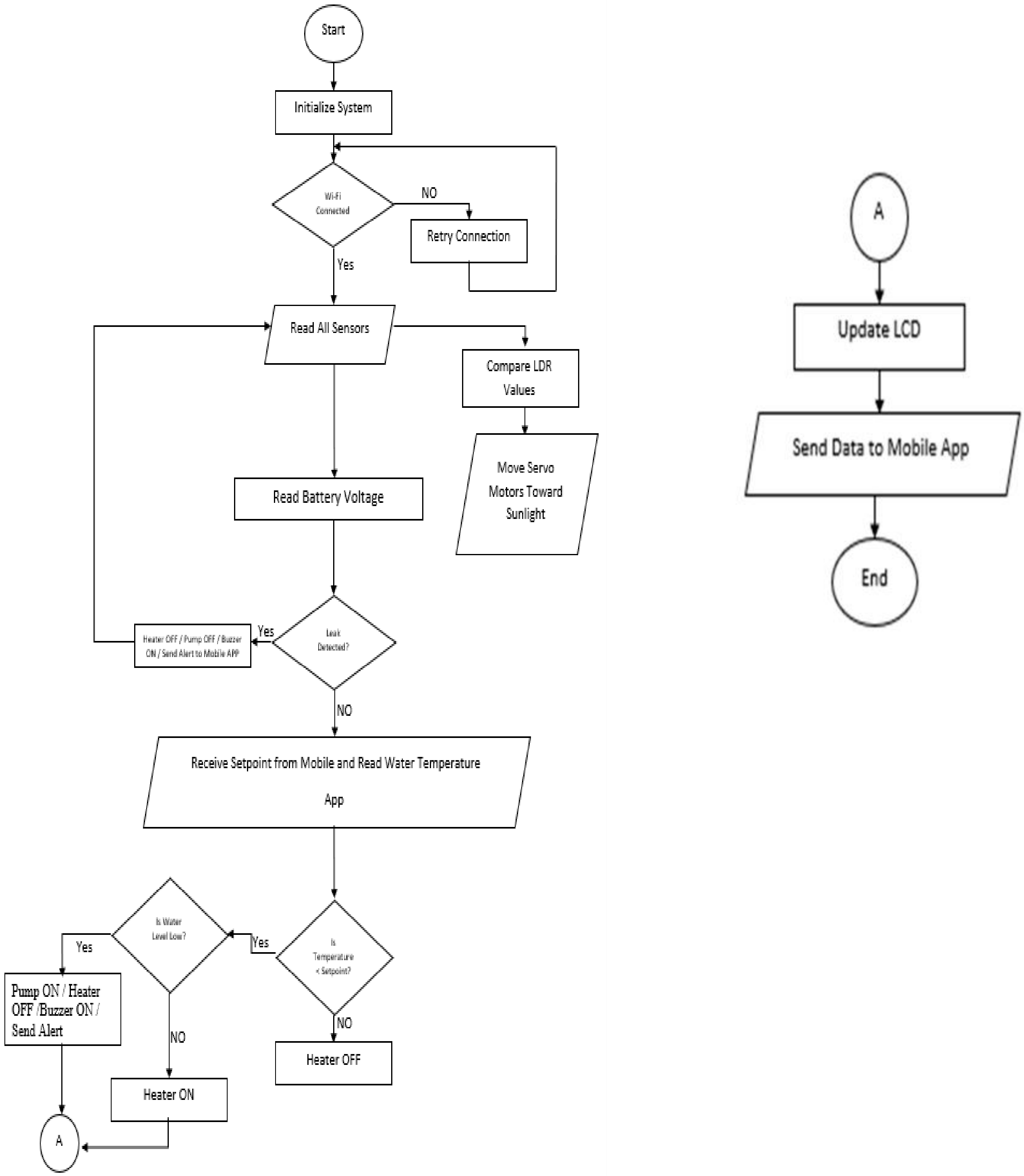

Figure 3.2: Flowchart of the Proposed System Control Strategy

3.4 Hardware Implementation

The proposed system is implemented using a hardware architecture centered around the ESP32 microcontroller, which acts as the main control unit. Sensors and actuators are interconnected according to the wiring diagram shown in Figure 3.3. This design ensures efficient integration of system components while maintaining stable power distribution and safe operation, based on established concepts such as solar tracking systems [1], temperature sensing [2], water management [3], and safety mechanisms [4].

The power system is based on a 12V battery charged by a solar panel. A DC-DC converter is used to step down the voltage to 5V for supplying the ESP32 and other low-voltage components. Sensors are powered directly from the ESP32 3.3V output, with all grounds connected in a common ground configuration to ensure signal stability and prevent measurement errors.

Four LDR sensors are connected to analog input pins (GPIO 32, 33, 34, 35) to detect light intensity from different directions. The DS18B20 temperature sensor is connected to GPIO 4 using the OneWire protocol, ensuring accurate temperature measurement [2]. The water level sensor is connected to GPIO 14, the leakage sensor to GPIO 39, and the voltage sensor to GPIO 36 for battery monitoring.

At the actuator level, two MG995 servo motors are connected to GPIO 25 and GPIO 26 to enable dual-axis solar tracking. High-power components such as the heater, water valve, and pump are controlled using driver circuits (e.g., BD243 transistors) connected to GPIO 19, GPIO 18, and GPIO 23, respectively, allowing safe switching of 12V loads. A buzzer is connected to GPIO 12 to provide audible alerts during fault conditions.

An I2C LCD is used for real-time data display, connected via SDA (GPIO 21) and SCL (GPIO 22), reducing wiring complexity. Low-voltage components such as servos, LCD, and pump are powered by 5V, while high-power loads such as the heater and valve operate at 12V.

Figure 3.3 illustrates the complete system wiring, including power distribution, sensor connections, and actuator interfacing, reflecting a well-integrated hardware design that ensures performance, safety, and ease of implementation.

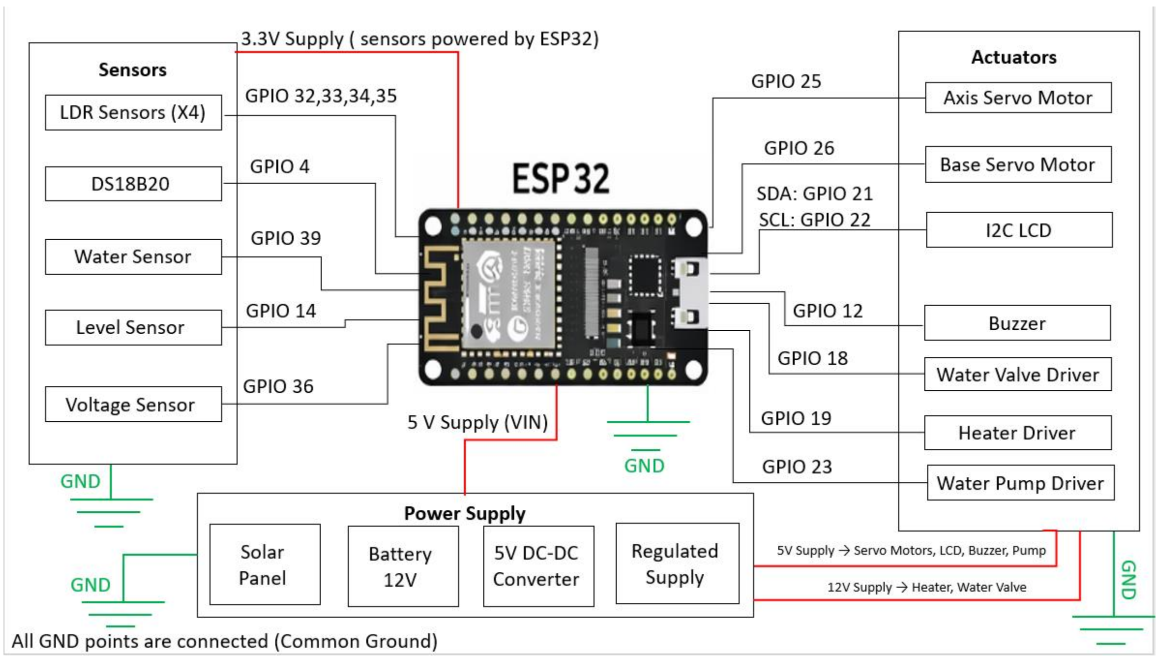

Figure 3.3: System Wiring Diagram

3.5 System Integration

The proposed system represents an integrated solution that combines hardware, software, and IoT communication within a unified operating environment based on the ESP32 microcontroller. The system software was developed using the Arduino IDE, utilizing several libraries such as Wi-Fi, OneWire, Dallas Temperature, LiquidCrystal_I2C, and ESP32Servo to ensure seamless integration between system components.

The system operates using a continuous loop-based mechanism, where sensor data are read, processed, and used for decision-making within the loop() function. The update interval is approximately 200 milliseconds, ensuring fast response and near real-time data updates.

In terms of communication, the ESP32 operates as a server within a local Wi-Fi network, establishing a direct connection with the mobile application using socket-based communication over port 80. This setup allows the system to receive the user-defined temperature setpoint and continuously transmit system data such as current temperature and battery voltage, as implemented in the mobile application code.

The mobile application was developed using Android Studio and provides an interactive interface that allows the user to input the desired temperature, monitor real-time system parameters, and receive alerts in case of abnormal conditions such as water leakage or low water level. Data reception is handled using a dedicated background thread to maintain application responsiveness and ensure stable performance.

The system operates in a fully integrated manner, where all subsystems—including solar tracking, thermal control, water management, and safety mechanisms—run concurrently without conflict or delay. This integration enables the system to respond dynamically and efficiently to changing operating conditions.

This design reflects a high level of integration between hardware, software, and communication systems, representing a practical implementation of smart energy systems as discussed in previous studies [1][2][3][4].

3.6 System Testing and Operational Scenarios

Since the proposed system is based on a real implementation rather than a theoretical simulation model, an experimental testing approach was adopted to evaluate system performance under different operating conditions. A set of operational scenarios was conducted to validate solar tracking efficiency, thermal control accuracy, and safety system responsiveness, based on the principles established in previous studies [1][2][3][4].

In the thermal control scenario, the system was tested under low initial temperature conditions. A temperature setpoint was entered via the mobile application, and it was observed that the heater automatically activated when the measured temperature was below the setpoint. The heater remained active until the desired temperature was reached, after which it turned off, demonstrating stable and effective thermal control performance [2].

For the leakage detection scenario, a water leakage condition was simulated. The system responded immediately by activating the buzzer and sending an alert to the mobile application, confirming the fast response and reliability of the conductivity-based leakage detection mechanism [4].

In the low water level scenario, the water level inside the tank was reduced. The system successfully triggered an alert and automatically activated the pump to refill the tank, while preventing unsafe operation, demonstrating the effectiveness of the water management system in avoiding dry-run conditions [3].

Additionally, the solar tracking system was verified to operate continuously, adjusting the panel orientation in response to variations in light intensity, thereby improving solar energy capture efficiency as described in solar tracking systems [1].

All tests were conducted using real measured sensor data rather than simulated datasets, ensuring high reliability and accuracy of the results. The system demonstrated stable and integrated performance, with all subsystems operating concurrently without conflict, confirming its effectiveness in achieving efficiency, safety, and real-time responsiveness.

3.7 Mathematical Modeling and Design Calculations

The thermal design of the system is based on fundamental thermodynamic relations [5], where the water heating process is described by:

$$\mathbf{Q = m * c * \Delta T}$$

Where

- Q is the required thermal energy
- m is the water mass
- c is the specific heat capacity
- ΔT is the temperature difference

The heating time is estimated using:

$$\mathbf{t = Q \backslash P}$$

Considering system losses, the actual heating time is given by:

$$\mathbf{tactual\ = Q \backslash \eta P}$$

where η represents system efficiency.

These equations describe the relationship between water volume, temperature variation, and heating power, and are derived from standard heat transfer principles as presented in fundamental engineering references [5].

3.8 Power and Electrical Analysis

The system is powered by a 12V battery, which serves as the main energy source. The voltage is distributed according to component requirements, where high-power loads such as the heater and solenoid valve operate at 12V, while a DC-DC converter is used to step down the voltage to 5V for the ESP32 and low-power components.

The overall energy consumption can be estimated by analyzing individual component loads. The ESP32 typically consumes between 80–260 mA, while actuators such as servos and pumps require higher current depending on operation.

The solar tracking system contributes to reducing overall energy consumption by improving solar energy capture efficiency, thereby reducing heater operating time [1].

This analysis confirms that the 12V battery is suitable for the system requirements, ensuring stable operation and efficient power management.

3.9 Communication Protocols

The system utilizes multiple communication protocols to ensure proper integration between its components. The I2C protocol is used to interface the LCD with the ESP32, reducing wiring complexity and simplifying hardware design.

The OneWire protocol is used for the DS18B20 temperature sensor, providing accurate temperature measurement using a single data line, making it suitable for precise thermal applications [2].

For external communication, the system relies on Wi-Fi technology, where the ESP32 operates as a server within a local network. Data exchange with the mobile application is achieved through socket-based communication, enabling real-time monitoring and remote control.

3.10 Chapter Conclusion

This chapter presented a comprehensive analysis and design of the proposed system, including the overall system architecture and key components. The control strategy and system operation were explained in detail, along with the hardware implementation and component interfacing.

System integration from both software and communication perspectives was also discussed, in addition to the testing methodology and operational scenarios. Furthermore, mathematical modeling and electrical analysis were introduced to provide a deeper understanding of system behavior and performance. Communication protocols were also presented to ensure proper interaction between system components.

Overall, this chapter demonstrated that the proposed system is based on an integrated design that balances efficiency, safety, and simplicity, providing a solid foundation for practical implementation and performance evaluation, which are discussed in the following chapter.

Chapter 4

Implementation and Testing

4.1 System Implementation

The proposed system was fully implemented in a real environment, where all electronic and mechanical components were assembled and interconnected according to the design specifications presented in the previous chapter. The system was tested as an integrated unit and demonstrated stable operation under real conditions.

Regarding the solar tracking system, the panel exhibited actual movement using servo motors, responding to variations in light intensity detected by LDR sensors. The system achieved an approximate tracking accuracy of around 80%, with minor fluctuations due to the analog nature of the measurements.

For the heating system, the heater is activated when a user-defined temperature setpoint is entered via the mobile application and operates until the desired temperature is reached, after which it turns off. If the temperature decreases afterward, the system does not automatically restart the heater; instead, it requires new user input. This reflects a user-triggered control approach.

In terms of communication, a stable connection was established between the ESP32 and the mobile application over a local Wi-Fi network, enabling real-time monitoring and control.

For safety and water management, both leakage and low water level scenarios were tested. In the case of leakage, the system activates the buzzer and sends an alert. When the water level drops, the system triggers an alert and automatically activates the pump to refill the tank, while preventing unsafe heater operation until normal conditions are restored. This confirms the effectiveness of both safety and water management mechanisms.

Finally, all subsystems—including solar tracking, water management, safety mechanisms, and the heating system under its defined control logic—operated simultaneously without conflict, demonstrating successful integration and stable system performance.

4.2 Experimental Results

The system performance was evaluated through a set of operational scenarios representing real working conditions, aiming to assess the efficiency of solar tracking, heating performance, and safety system responsiveness.

For the heating system, the system demonstrated the ability to increase the water temperature from an initial value of approximately 22∘C to a desired setpoint of 40∘C within an estimated time range of **60 to 90 minutes**, depending on water volume and operating conditions. This behavior is consistent with fundamental heat transfer principles [5].

Regarding solar tracking, the system showed noticeable improvement in panel orientation compared to a fixed position. The efficiency gain can be estimated within the range of **25% to 30%**, which aligns with reported values for dual-axis tracking systems [1].

In terms of battery performance, the system operates at approximately **12V** under full charge conditions, with a gradual voltage drop under load reaching approximately **11.2V to 11.6V**, which is typical for battery-powered systems.

The safety mechanisms demonstrated fast response times, with leakage detection and low water level conditions triggering system alerts within approximately **1 to 2 seconds**. In the case of low water level, the pump is automatically activated to refill the tank until normal conditions are restored.

For communication performance, the system achieved near real-time data updates, with response times ranging between **0.5 to 1 second**, ensuring effective monitoring and control via the mobile application.

Overall, the system performance falls within expected operational ranges, demonstrating reliable functionality, fast response, and effective integration of all subsystems.

Table 4.1: Experimental Results of the Proposed System

Scenario	Initial Conditions	Action	Result	Response Time / Performance
Water Heating	≈22°C → 40°C	Heater activation	Reached target temperature	≈60–90 minutes
Solar Tracking	Variable light	Servo movement	Improved panel orientation	≈25%–30% efficiency gain
Battery Voltage	≈12V (fully charged)	System operation	Gradual voltage drop under load	≈11.2–11.6V
Water Leakage	Leakage detected	Safety activation	Alarm + notification	≈1–2 seconds
Low Water Level	Reduced water level	Pump activation	Automatic tank refilling	≈1–3 seconds
Communication	System connected	Data exchange	Real-time updates	≈0.5–1 second

4.3 Performance Evaluation (KPIs)

The system performance was evaluated using a set of Key Performance Indicators (KPIs) to assess the achievement of design objectives, including tracking accuracy, system response time, thermal control performance, communication efficiency, and overall system reliability.

For solar tracking, the system achieved an approximate tracking accuracy of **±5°**, with a servo response time of **1–2 seconds**, which is acceptable for low-cost LDR-based systems. This resulted in an estimated energy gain of **25%–30%** compared to fixed panels [1].

In the heating system, the desired temperature was successfully achieved, with a thermal stability of approximately **±2°C**, which is reasonable for ON/OFF control systems [2].

In terms of monitoring and communication, the system demonstrated synchronized updates on both the LCD and mobile application with a response time of approximately **0.5 to 1 second**, enabling effective real-time monitoring.

For safety mechanisms, both leakage detection and low water level response times were within **1–2 seconds**, with immediate alarm activation and automatic pump operation when required.

Regarding energy efficiency, the solar tracking system contributed to reducing heater operating time, resulting in an estimated reduction in energy consumption of **15%–25%** compared to non-tracking systems.

Finally, the system demonstrated high reliability, with all subsystems operating concurrently without failures or conflicts, confirming successful integration and stable operation.

Table 4.2: System Performance Indicators (KPIs)

Indicator	Approximate Value	Evaluation
Solar Tracking Accuracy	±5°	Good
Servo Response Time	1–2 seconds	Fast
Energy Efficiency Improvement	25%–30%	Significant
Thermal Control Accuracy	±2°C	Stable
Data Update Rate	0.5–1 second	Near real-time
Leakage Detection Response	1–2 seconds	Fast
Low Water Level Response	1–3 seconds	Effective
Energy Consumption Reduction	15%–25%	Good
System Reliability	Continuous operation without failure	High

4.4 Critical Analysis of the Project

This section presents a critical evaluation of the implemented system by analyzing its strengths, limitations, and overall performance in relation to the defined KPIs and experimental results.

In terms of strengths, the system demonstrated effective integration of all subsystems, including solar tracking, thermal control, water management, and IoT-based monitoring. All components operated simultaneously without conflict, which is reflected in the performance indicators such as fast response times (1–2 seconds) for leakage and low water level detection, near real-time data updates (0.5–1 second), and an estimated energy efficiency improvement of 25%–30% compared to fixed systems [1]. The system also achieved the desired temperature with acceptable thermal stability (±2°C), confirming the effectiveness of the overall design.

Regarding limitations, the solar tracking accuracy did not reach ideal levels, with an approximate deviation of ±5°, which is expected in LDR-based systems due to sensitivity to environmental conditions such as light variation and reflections [1]. Additionally, the heating system operates using an ON/OFF logic triggered by user input, meaning it does not automatically maintain temperature once it drops, which limits full automation. The system performance is also dependent on battery capacity and solar charging conditions, which may affect operation under low irradiance conditions.

Compared to advanced systems in the literature, the proposed system offers advantages in terms of simplicity, cost-effectiveness, and ease of implementation, while integrating multiple functionalities into a single platform. However, more advanced systems may employ sophisticated control algorithms (e.g., adaptive control or AI-based optimization), achieving higher precision and efficiency at the cost of increased complexity.

For future work, the system can be enhanced by integrating cloud-based platforms for extended monitoring and control, applying artificial intelligence techniques for optimized energy management, implementing predictive maintenance for early fault detection, developing intelligent scheduling based on user behavior and environmental conditions, and enabling smart home integration. These improvements would significantly enhance system performance and scalability.

Overall, the system achieves a balanced trade-off between performance, cost, and simplicity, with strong potential for future development toward more intelligent and efficient operation.

4.5 Cost Analysis and Final Evaluation

A preliminary cost analysis of the proposed system was conducted based on approximate component prices in the local market (Damascus) to evaluate the economic feasibility. Table 4.3 presents the estimated costs of the main components, including price ranges to account for market variability and exchange rate fluctuations.

Table 4.3: Estimated Cost of System Components (USD)

Component	Quantity	Estimated Cost (USD)
ESP32	1	12 – 18
Servo Motors (MG995)	2	18 – 30
LDR Sensors	4	1 – 3
DS18B20	1	2 – 5
Water Level + Leak Sensors	2	3 – 7
Water Pump	1	8 – 15
Solenoid Valve	1	10 – 18
Heater Element	1	10 – 20
LCD (I2C)	1	5 – 10
Battery (12V)	1	25 – 50
Misc. (Wiring, Drivers, Frame)	—	15 – 35
Total Estimated Cost	—	**109 – 211 USD**

The system's initial cost is moderate, particularly due to the inclusion of tracking components such as motors and mechanical structures. However, long-term savings are achieved through improved energy efficiency and reduced number of solar panels compared to fixed systems [1].

The system demonstrated stable and effective performance, meeting all primary objectives including solar tracking, heating, safety, and monitoring. It also offers strong potential for future development to enhance efficiency and reduce costs using advanced techniques, as discussed in the future work section.

4.6 Chapter Conclusion

This chapter presented the practical implementation and performance evaluation of the proposed system. The system was successfully built and operated as an integrated unit under real conditions, demonstrating its ability to achieve the main objectives, including solar tracking, thermal control, and safety functions.

The experimental results were analyzed across different operating scenarios, confirming the system's effectiveness and reliability. Performance was further evaluated using Key Performance Indicators (KPIs), which demonstrated satisfactory response time, accuracy, and overall system stability.

A critical analysis was also conducted to highlight both the strengths and limitations of the system, reflecting a comprehensive understanding of its behavior and performance. In addition, a cost analysis was provided to assess the economic feasibility of the system.

Overall, this chapter confirms the successful implementation and operation of the proposed system, providing a solid basis for the final conclusions and future recommendations presented in the next chapter.

Chapter 5

Conclusion and Recommendations

5.1 Project Summary and Key Achievements

This project presents an integrated smart system that combines dual-axis solar tracking and water heating, along with safety and monitoring functions, within a single platform based on the ESP32 microcontroller. The system aims to improve solar energy utilization efficiency while ensuring safe and reliable operation.

All main project objectives were successfully achieved. The solar tracking system effectively improves panel orientation and energy capture efficiency, while the heating system is capable of reaching the desired temperature. In addition, a Wi-Fi-based communication system was implemented to enable remote monitoring and control, along with safety mechanisms including water leakage detection and water level monitoring.

The proposed system is distinguished by integrating multiple functions (tracking, heating, monitoring, and safety) into a relatively simple and cost-effective design compared to more advanced systems, while maintaining acceptable performance, in line with concepts presented in previous studies [1][2][3][4].

The project demonstrates that it is possible to develop a practical and integrated system that balances efficiency, responsiveness, and ease of implementation, confirming the success of the design and its potential for real-world applications and future improvements.

5.2 Challenges and Difficulties

The implementation of the project involved several technical and practical challenges, which are typical in multi-component integrated systems. From a technical perspective, one of the main challenges was related to the accuracy of the solar tracking system. The use of LDR sensors, which rely on analog measurements, makes the system sensitive to environmental factors such as light variation and reflections, resulting in minor fluctuations in tracking accuracy, as reported in solar tracking studies [1].

In the thermal control system, an ON/OFF control approach was adopted due to its simplicity and reliability. However, this method may not maintain perfectly stable temperature under all conditions, which is expected for this type of control system [2].

On the practical side, challenges were encountered in component wiring and system organization, especially due to the integration of multiple sensors and actuators. Proper wiring and grounding were essential to ensure signal stability and avoid measurement errors. Additionally, managing different voltage levels (3.3V, 5V, and 12V) required careful design to ensure safe and stable operation.

Furthermore, the availability of components in the local market and price variations posed organizational challenges, along with limited time for extensive experimental testing, which led to focusing primarily on core system functionalities.

Overall, these challenges contributed to a deeper understanding of the system and helped refine its design, leading to a balanced solution that combines performance, simplicity, and practical feasibility.

5.3 Future Work

The proposed system can be significantly enhanced through several technical improvements aimed at increasing performance, accuracy, and overall intelligence. One of the main areas of improvement is the solar tracking system, where higher-precision sensors or improved signal processing algorithms can be used to reduce environmental noise effects and enhance tracking accuracy.

The thermal control system can also be improved by replacing the basic ON/OFF control with more advanced control algorithms such as Proportional-Integral-Derivative (PID) control, which can provide better temperature stability and reduce fluctuations around the desired setpoint.

From a communication perspective, the system can be integrated with cloud-based platforms to enable extended remote monitoring and control, as well as data logging and long-term analysis. Artificial intelligence techniques can also be applied to optimize energy consumption and support intelligent decision-making based on real operating data.

Another important enhancement is the implementation of predictive maintenance, where sensor data can be analyzed to detect potential faults before they occur, thereby improving system reliability. Additionally, smart scheduling can be introduced based on user behavior and environmental conditions such as solar radiation and ambient temperature.

Finally, the system can be integrated with smart home platforms, allowing it to operate within a broader intelligent environment and enabling centralized control.

Overall, these improvements provide a clear pathway for transforming the system from a functional prototype into a more advanced, intelligent, and scalable solution.

5.4 Recommendations

Based on the experience gained from this project, it is recommended that students working on similar systems focus on proper planning and system design, particularly in terms of component organization and wiring, as this plays a critical role in ensuring system stability and reducing operational errors. It is also advisable to test each subsystem independently before full integration, allowing for easier troubleshooting and performance optimization.

From a practical perspective, selecting reliable and suitable components is essential, especially for analog-based systems such as LDR sensors, as component quality directly affects system accuracy. Proper energy management and battery selection are also crucial to ensure continuous and stable operation, particularly in solar-powered systems.

From a technical standpoint, implementing more advanced control algorithms in future stages can significantly improve system stability and efficiency. Additionally, proper documentation of all development stages is highly recommended to support future improvements.

Overall, this project demonstrates that success is not only determined by the idea itself, but by effective implementation, practical understanding, and the ability to handle real-world challenges.

5.5 Chapter Conclusion

This chapter concluded the overall project by summarizing the key achievements, discussing the challenges encountered during implementation, and presenting future development opportunities and practical recommendations. It reflects a comprehensive understanding of all project stages, from design to implementation and evaluation.

In general, the project demonstrates the feasibility of developing an integrated smart system that combines solar tracking, thermal control, and safety mechanisms within a single platform, while maintaining a balance between performance, cost, and simplicity. It also highlights the importance of such systems in supporting renewable energy utilization and smart system applications in real-life scenarios.

This work provides a solid foundation for future research and development, whether through improving system accuracy, enhancing control strategies, or expanding application scope. Therefore, the project concludes with a practical and scientific framework that opens the door for further innovation and advancement in this field.

Appendices

The appendices include supporting materials for the report, including source codes, and the user manual, in order to fully document the design and implementation stages.

Appendix A: Source Code

This appendix includes the source codes developed for the system implementation, including the ESP32 firmware and the mobile application code.

A.1 ESP32 Main Control Code

The following code represents the main ESP32 firmware used to control solar tracking, heating, safety mechanisms, LCD display, and Wi-Fi communication.

Listing A.1: ESP32 Main Control Code

```
#include <WiFi.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#include <OneWire.h>

#include <DallasTemperature.h>

#include <ESP32Servo.h>


const char* ssid = "YAZAN";

const char* password = "***************";


IPAddress local_IP(192, 168, 1, 33);

IPAddress gateway(192, 168, 1, 1);
```

```
IPAddress subnet(255, 255, 255, 0);

WiFiServer server(80);
WiFiClient client;

#define ONE_WIRE_BUS 4

String data_rec;
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);
LiquidCrystal_I2C lcd(0x27,16,2);

Servo myservo_1, myservo_2;

int pos_1 = 85, pos_2 = 180;
int data_1, data_2, data_3, data_4;
int movee = 0, wanted = 0;

float volt, temp;

void setup() {
```

```
Serial.begin(9600);

if (!WiFi.config(local_IP, gateway, subnet)) {
  Serial.println("STA Failed to configure");
}

WiFi.begin(ssid, password);

Serial.print("Connecting to WiFi");
while (WiFi.status() != WL_CONNECTED) {
  delay(500);
  Serial.print(".");
}

Serial.println("\nConnected!");
Serial.println(WiFi.localIP());

server.begin();

pinMode(14, INPUT_PULLUP);
pinMode(12, OUTPUT);
pinMode(18, OUTPUT);
```

```
  pinMode(19, OUTPUT);
  pinMode(23, OUTPUT);
  analogSetAttenuation(ADC_11db);

  sensors.begin();
  Wire.begin(21,22);
  lcd.begin();
  lcd.backlight();

  myservo_1.attach(25);
  myservo_1.write(pos_1);

  myservo_2.attach(26);
  myservo_2.write(pos_2);
}

void loop() {

  WiFiClient newClient = server.available();
  if (newClient) {
    client = newClient;
    Serial.println("Client connected");
```

```
}

if (client && client.connected() && client.available()) {
  data_rec = client.readStringUntil('\n');
  wanted = data_rec.toInt();
  lcd.clear();
}

if ((digitalRead(14) == HIGH) && (digitalRead(12) == LOW)) {
  if (client && client.connected()) client.println("c");
  digitalWrite(12, HIGH);
  digitalWrite(18, HIGH);
  digitalWrite(23, HIGH);
}

if ((digitalRead(14) == LOW) && (digitalRead(23) == HIGH)) {
  digitalWrite(23, LOW);
}

int watterr = analogRead(39);

if ((watterr > 500) && (digitalRead(12) == LOW)) {
```

```
    if (client && client.connected()) client.println("a");

    digitalWrite(18, HIGH);

    digitalWrite(12, HIGH);

  }


  if ((watterr < 500) && (digitalRead(12) == HIGH) && (digitalRead(14) ==
LOW)) {

    if (client && client.connected()) client.println("d");

    digitalWrite(12, LOW);

    digitalWrite(18, LOW);

  }


  int adcValue = analogRead(36);

  volt = (float)adcValue * 3.3 / 4095.0;

  volt = volt * 3.9;


  sensors.requestTemperatures();

  temp = sensors.getTempCByIndex(0);


  if (client && client.connected()) {

    client.println("V");

    client.println(volt);

    client.println("T");
```

```
  client.println(temp);
}

delay(200);

if (temp >= wanted) {
  digitalWrite(19, LOW);
} else {
  digitalWrite(19, HIGH);
}

lcd.setCursor(0,0);
lcd.print("now");
lcd.setCursor(0,1);
lcd.print(temp);

lcd.setCursor(6,0);
lcd.print("want");
lcd.setCursor(6,1);
lcd.print(wanted);

lcd.setCursor(11,0);
```

```
lcd.print("volt");
lcd.setCursor(11,1);
lcd.print(volt);

data_1 = analogRead(33);
data_2 = analogRead(35);

if (data_1 > 2800) myservo_1.write(70);
if (data_2 > 2800) myservo_1.write(145);

data_3 = analogRead(32);
data_4 = analogRead(34);

if ((data_3 > 2500) || (data_4 > 2500)) {
  if (movee == 0) {
    movee = 1;
    pos_2 = (pos_2 == 180) ? 80 : 180;
    myservo_2.write(pos_2);
  }
} else {
  movee = 0;
}}
```

A.2 Mobile Application User Interface Code

The following code represents the user interface layout of the mobile application developed using Android Studio. It allows the user to set the desired temperature, monitor current temperature and voltage, and control the system via Wi-Fi connection.

Listing A.2: Mobile Application UI Code

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#0B1C2C"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/title"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="SMART CONTROL PANEL"
        android:textColor="#00E5FF"
        android:textStyle="bold"
        android:textSize="22sp"
        app:layout_constraintTop_toTopOf="parent"
```

```
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="24dp"/>

<ToggleButton
    android:id="@+id/conn"
    android:layout_width="70dp"
    android:layout_height="70dp"
    android:background="@drawable/wifioff"
    android:textOn=""
    android:textOff=""
    android:onClick="onToggleClicked_1"
    app:layout_constraintTop_toBottomOf="@id/title"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="16dp"
    android:layout_marginEnd="16dp"/>

<androidx.cardview.widget.CardView
    android:id="@+id/card1"
    android:layout_width="0dp"
    android:layout_height="100dp"
    app:cardCornerRadius="16dp"
    app:cardBackgroundColor="#142B3D"
    app:layout_constraintTop_toBottomOf="@id/conn"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
```

```
android:layout_margin="12dp">

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="horizontal"
    android:gravity="center_vertical"
    android:padding="16dp">

    <TextView
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="TEMP SET"
        android:textColor="#FFFFFF"
        android:textSize="18sp" />

    <EditText
        android:id="@+id/EditTexttttt"
        android:layout_width="80dp"
        android:layout_height="50dp"
        android:background="#1E3A52"
        android:gravity="center"
        android:text="00"
        android:textColor="#00E5FF"
        android:textSize="22sp"
```

```xml
            android:inputType="number"
            android:maxLength="3"
            android:cursorVisible="true" />

    </LinearLayout>
</androidx.cardview.widget.CardView>

<androidx.cardview.widget.CardView
    android:id="@+id/card2"
    android:layout_width="0dp"
    android:layout_height="100dp"
    app:cardCornerRadius="16dp"
    app:cardBackgroundColor="#142B3D"
    app:layout_constraintTop_toBottomOf="@id/card1"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_margin="12dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="horizontal"
        android:gravity="center_vertical"
        android:padding="16dp">

        <TextView
```

```xml
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="CURRENT TEMP"
            android:textColor="#FFFFFF"
            android:textSize="18sp" />

        <TextView
            android:id="@+id/noww"
            android:layout_width="80dp"
            android:layout_height="50dp"
            android:background="#1E3A52"
            android:gravity="center"
            android:text="30"
            android:textColor="#00E5FF"
            android:textSize="22sp" />

    </LinearLayout>
</androidx.cardview.widget.CardView>

<androidx.cardview.widget.CardView
    android:id="@+id/card3"
    android:layout_width="0dp"
    android:layout_height="100dp"
    app:cardCornerRadius="16dp"
    app:cardBackgroundColor="#142B3D"
```

```
app:layout_constraintTop_toBottomOf="@id/card2"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"
android:layout_margin="12dp">

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="horizontal"
    android:gravity="center_vertical"
    android:padding="16dp">

    <TextView
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="VOLTAGE"
        android:textColor="#FFFFFF"
        android:textSize="18sp" />

    <TextView
        android:id="@+id/voltt"
        android:layout_width="80dp"
        android:layout_height="50dp"
        android:background="#1E3A52"
        android:gravity="center"
```

```
            android:text="12"
            android:textColor="#00E5FF"
            android:textSize="22sp" />

    </LinearLayout>
</androidx.cardview.widget.CardView>

<Button
    android:id="@+id/sendd"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:text="SEND DATA"
    android:textStyle="bold"
    android:textSize="18sp"
    android:textColor="#FFFFFF"
    android:backgroundTint="#00E5FF"
    app:layout_constraintTop_toBottomOf="@id/card3"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_margin="16dp"/>

<TextView
    android:id="@+id/showww"
    android:layout_width="0dp"
    android:layout_height="50dp"
    android:text="SYSTEM READY"
```

```
        android:gravity="center"
        android:textColor="#00E5FF"
        android:textSize="18sp"
        android:textStyle="bold"
        app:layout_constraintTop_toBottomOf="@id/sendd"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_margin="12dp"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

A.3 Mobile Application Logic & Communication Code

The following code represents the main logic of the mobile application, including Wi-Fi communication with the ESP32, data transmission, and real-time data reception.

Listing A.3: Mobile Application Logic and Communication Code

```
package com.example.myapplication;

import android.os.Bundle;

import android.os.Handler;

import android.os.Looper;

import android.view.View;

import android.widget.Button;
```

```java
import android.widget.EditText;

import android.widget.TextView;

import android.widget.ToggleButton;

import androidx.appcompat.app.AppCompatActivity;


import java.io.BufferedReader;

import java.io.IOException;

import java.io.InputStream;

import java.io.InputStreamReader;

import java.io.OutputStream;

import java.net.InetSocketAddress;

import java.net.Socket;


public class MainActivity extends AppCompatActivity {


    String SERVER_IP = "192.168.1.33";

    int SERVER_PORT = 80;
```

```
Socket socket;

OutputStream outputStream;

InputStream inputStream;



Thread workerThread;

volatile boolean stopWorker;



EditText EditTextttt;

TextView nowww, volttt, showww;

ToggleButton connn;



int tt = 0, vv = 0;



@Override

protected void onCreate(Bundle savedInstanceState) {

    super.onCreate(savedInstanceState);

    setContentView(R.layout.activity_main);
```

```
Button senddd = findViewById(R.id.sendd);

connn    = findViewById(R.id.conn);

showww   = findViewById(R.id.showww);

nowww    = findViewById(R.id.noww);

volttt   = findViewById(R.id.voltt);

EditTextttt = findViewById(R.id.EditTextttt);



senddd.setOnClickListener(v -> {



   if (socket == null || !socket.isConnected() || outputStream == null) {

      showww.setText("Not Connected ❌");

      return;

   }



   String msg = EditTextttt.getText().toString().trim();



   if (msg.isEmpty()) {

      showww.setText("Enter a value first ⚠️");
```

```
            return;
        }

        new Thread(() -> {
            try {
                byte[] data = (msg + "\n").getBytes("UTF-8");
                outputStream.write(data);
                outputStream.flush();

                runOnUiThread(() -> showww.setText("Sent: " + msg + " ✅"));

            } catch (IOException e) {
                e.printStackTrace();
                runOnUiThread(() -> showww.setText("Send Error: " + e.getMessage() + " ⚠️"));
            }
        }).start();
```

```
        });

    }



    public void onToggleClicked_1(View v) {

        if (connn.isChecked()) {

            new Thread(() -> {

                try {

                    socket = new Socket();

                    socket.connect(new                    InetSocketAddress(SERVER_IP,
SERVER_PORT), 3000);

                    outputStream = socket.getOutputStream();

                    inputStream  = socket.getInputStream();



                    runOnUiThread(() -> {

                        connn.setBackgroundResource(R.drawable.wifion);

                        showww.setText("Connected ✅");

                    });
```

```
            beginListenForData();


        } catch (Exception e) {
            e.printStackTrace();
            runOnUiThread(() -> {
                connn.setChecked(false);
                connn.setBackgroundResource(R.drawable.wifioff);
                showww.setText("Connection Failed: " + e.getMessage());
            });
        }
    }).start();


} else {
    new Thread(() -> {
        try {
            closeConnection();
        } catch (IOException e) {
            e.printStackTrace();
```

```
            }
            runOnUiThread(() -> {
                showww.setText("Disconnected");
                connn.setBackgroundResource(R.drawable.wifioff);
            });
        }).start();
    }
}

void beginListenForData() {
    Handler handler = new Handler(Looper.getMainLooper());
    stopWorker = false;

    workerThread = new Thread(() -> {
        try {
            BufferedReader reader = new BufferedReader(
                    new InputStreamReader(inputStream, "UTF-8"));
```

```
while (!stopWorker && socket != null && socket.isConnected()) {

    String data = reader.readLine();

    if (data == null) break;



    String finalData = data.trim();



    handler.post(() -> {

        if (finalData.equals("V")) {

            vv = 1;

        } else if (finalData.equals("T")) {

            tt = 1;

        } else if (vv == 1) {

            vv = 0;

            volttt.setText(finalData);

        } else if (tt == 1) {

            tt = 0;

            nowww.setText(finalData);

        } else if (finalData.equals("a")) {
```

```
                showww.setText("Water leak 🚨");
            } else if (finalData.equals("c")) {
                showww.setText("Low water level ⚠");
            } else if (finalData.equals("d")) {
                showww.setText("--------------------");
            }
        });
    }

} catch (IOException e) {
    runOnUiThread(() -> {
        showww.setText("Disconnected ⚠");
        connn.setChecked(false);
        connn.setBackgroundResource(R.drawable.wifioff);
    });
}
});
```

```
        workerThread.start();
    }

    void closeConnection() throws IOException {
        stopWorker = true;

        if (workerThread != null) {
            workerThread.interrupt();
            workerThread = null;
        }
        try { if (outputStream != null) outputStream.close(); } catch (IOException ignored) {}
        try { if (inputStream  != null) inputStream.close();  } catch (IOException ignored) {}
        try { if (socket       != null) socket.close();       } catch (IOException ignored) {}

        outputStream = null;
        inputStream  = null;
```

```
        socket       = null;
    }


    @Override
    protected void onDestroy() {
        super.onDestroy();
        new Thread(() -> {
            try { closeConnection(); } catch (IOException ignored) {}
        }).start();
    }
}
```

Appendix B: User Manual

This appendix provides the basic user manual for operating the system, including startup steps, mobile application connection, temperature setting, and safety monitoring.

B.1 System Startup

1. Ensure that the 12V battery is properly connected.
2. Verify that all sensors and actuators are correctly wired.
3. Power on the system and wait for the ESP32 to connect to the Wi-Fi network.

4. Confirm that system data appears on the LCD display.

B.2 Mobile Application Connection

1. Open the mobile application.
2. Press the Wi-Fi connection button.
3. Once connected, the connection status will be displayed.
4. The application will begin showing real-time temperature and battery voltage.

B.3 Temperature Setting

1. Enter the desired temperature in the **TEMP SET** field.
2. Press the **SEND DATA** button.
3. The value will be sent to the ESP32.
4. The heater will operate until the target temperature is reached, then it will stop.

B.4 Safety Monitoring

- In case of water leakage, an alarm will be activated and a warning message will appear in the application.
- In case of low water level, an alarm will be triggered and the pump will automatically refill the tank.
- The system continuously displays temperature and voltage on both the LCD and the mobile application.

B.5 Shutdown Procedure

1. Disconnect from the mobile application.
2. Turn off the system by disconnecting the 12V power supply.

3. Ensure that the heater and pump are turned off before shutting down the system.

References

[1] J. Rizk and Y. Chaiko,
“Solar Tracking System: More Efficient Use of Solar Panels,”
World Academy of Science, Engineering and Technology, vol. 41, pp. 313–315, 2008.

[2] Maxim Integrated,
“DS18B20 Programmable Resolution 1-Wire Digital Thermometer Datasheet,”
2015.

[3] S. Kumar and A. Singh,
“Design and Implementation of Automatic Water Level Controller,”
International Journal of Engineering Research & Technology (IJERT), vol. 3, no. 9, pp. 100–104, 2014.

[4] A. Sharma,
“Water Leakage Detection System Using Conductivity Method,”
International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering, vol. 5, no. 6, pp. 456–460, 2016.

[5] Y. A. Çengel and M. A. Boles,
Thermodynamics: An Engineering Approach, 8th ed.
New York: McGraw-Hill, 2015.